Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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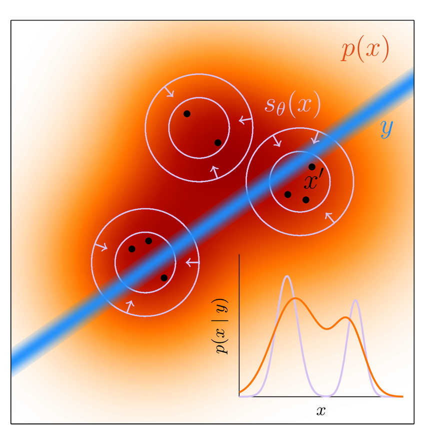
The question

Generative emulators promise **cheap surrogates** for expensive cosmological simulations — but an emulator that **memorizes** its training set can give a biased prior over fields. We train diffusion models on CAMELS HI maps and ask:

1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?



If the learned prior collapses onto training islands, a downstream posterior can become biased even when the likelihood calculation is sharp.

Data & models

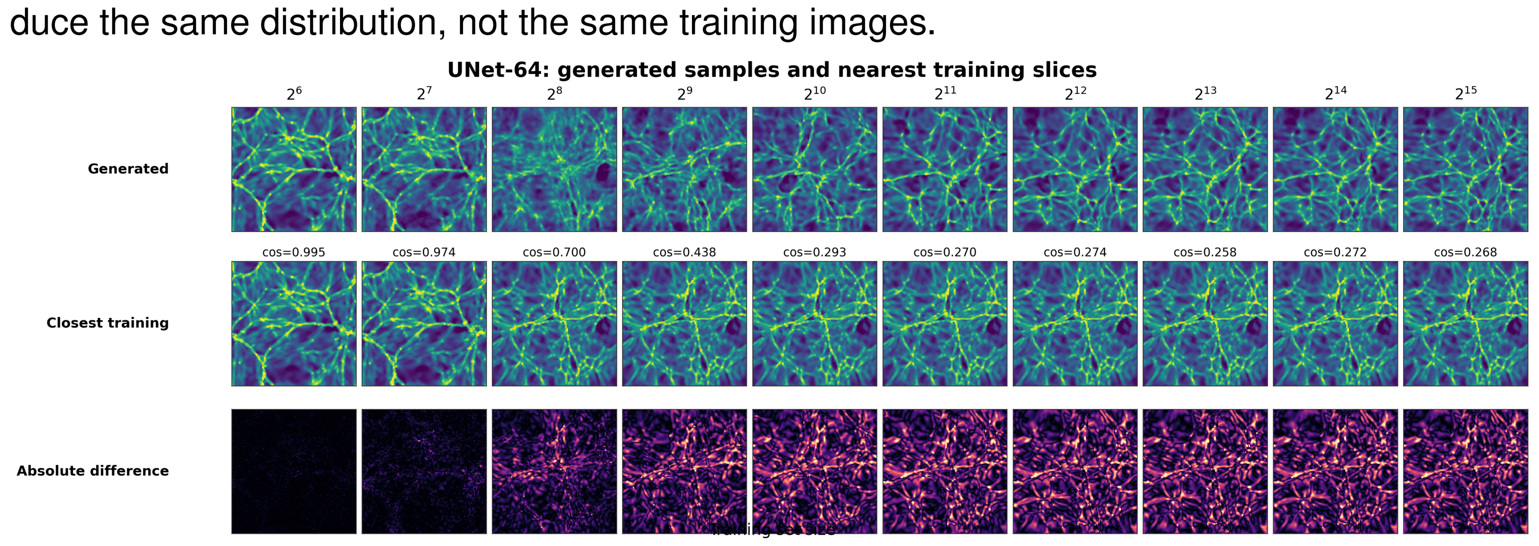
- **CAMELS** (IllustrisTNG) HI fields $\rightarrow$ 128×128 slices; 6 parameters Ω_m, σ_8, A_{SN1}, A_{AGN1}, A_{SN2}, A_{AGN2}.
- **Diffusion objective**: train a UNet denoiser with $x_t = \sqrt{\bar\alpha_t}x_0 + \sqrt{1-\bar\alpha_t}\epsilon$ and $v_t = \sqrt{\bar\alpha_t}\epsilon - \sqrt{1-\bar\alpha_t}x_0$; minimize $\mathcal{L}_{\mathrm{diff}} = \mathbb{E}\, w(t)\|f_\theta(x_t, t, c) - v_t\|_2^2$; $c = \boldsymbol{\theta}$ for conditional runs.
- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

Detecting memorization

For each generated sample x_j, find its nearest training field in pixel, PCA, and SSCD embedding space:

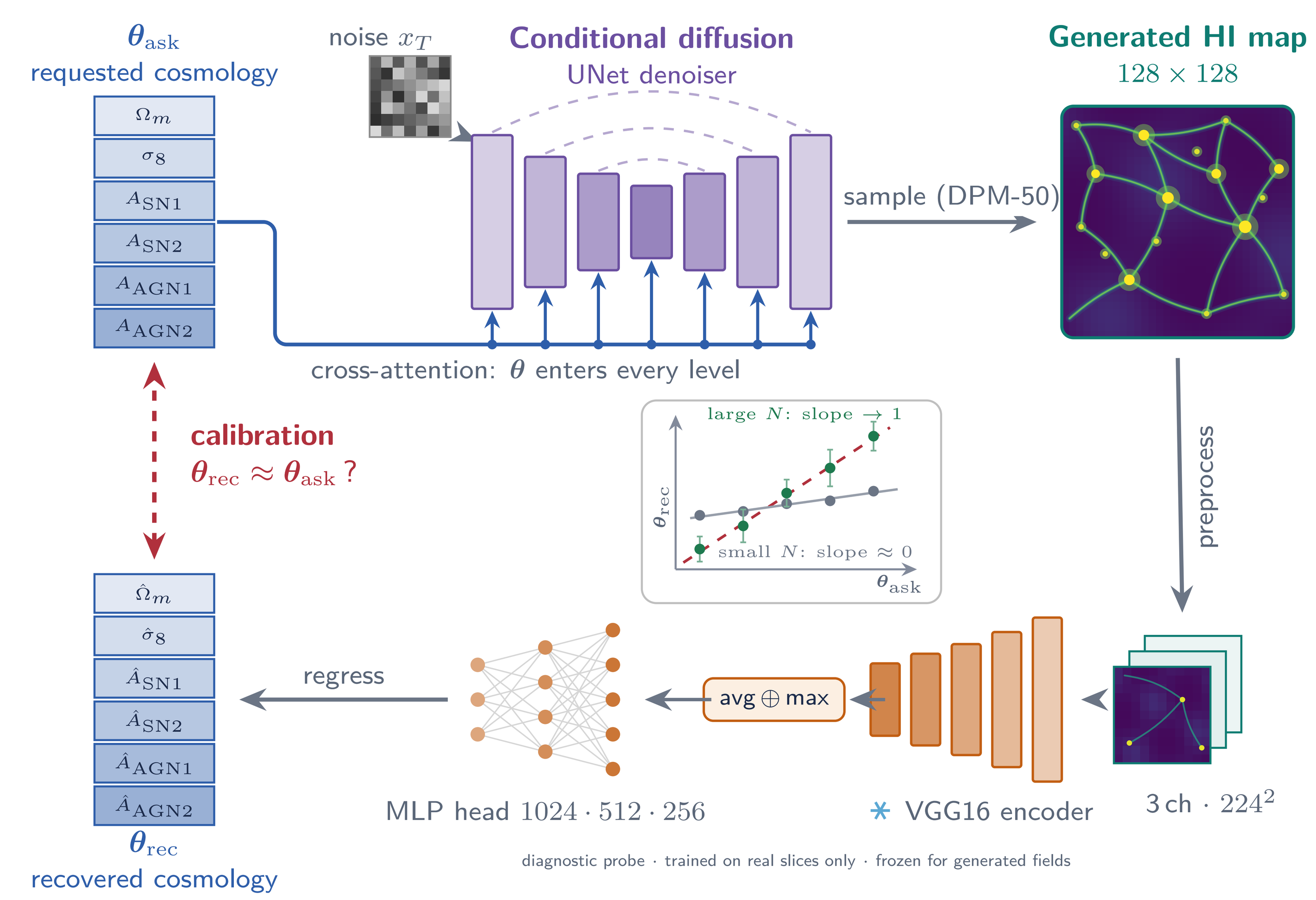
$$s_j = \max_i \mathrm{sim}(x_j, y_i), \qquad \mathrm{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

with τ calibrated from leave-one-out train–train similarities. Independent models should reproduce the same distribution, not the same training images.



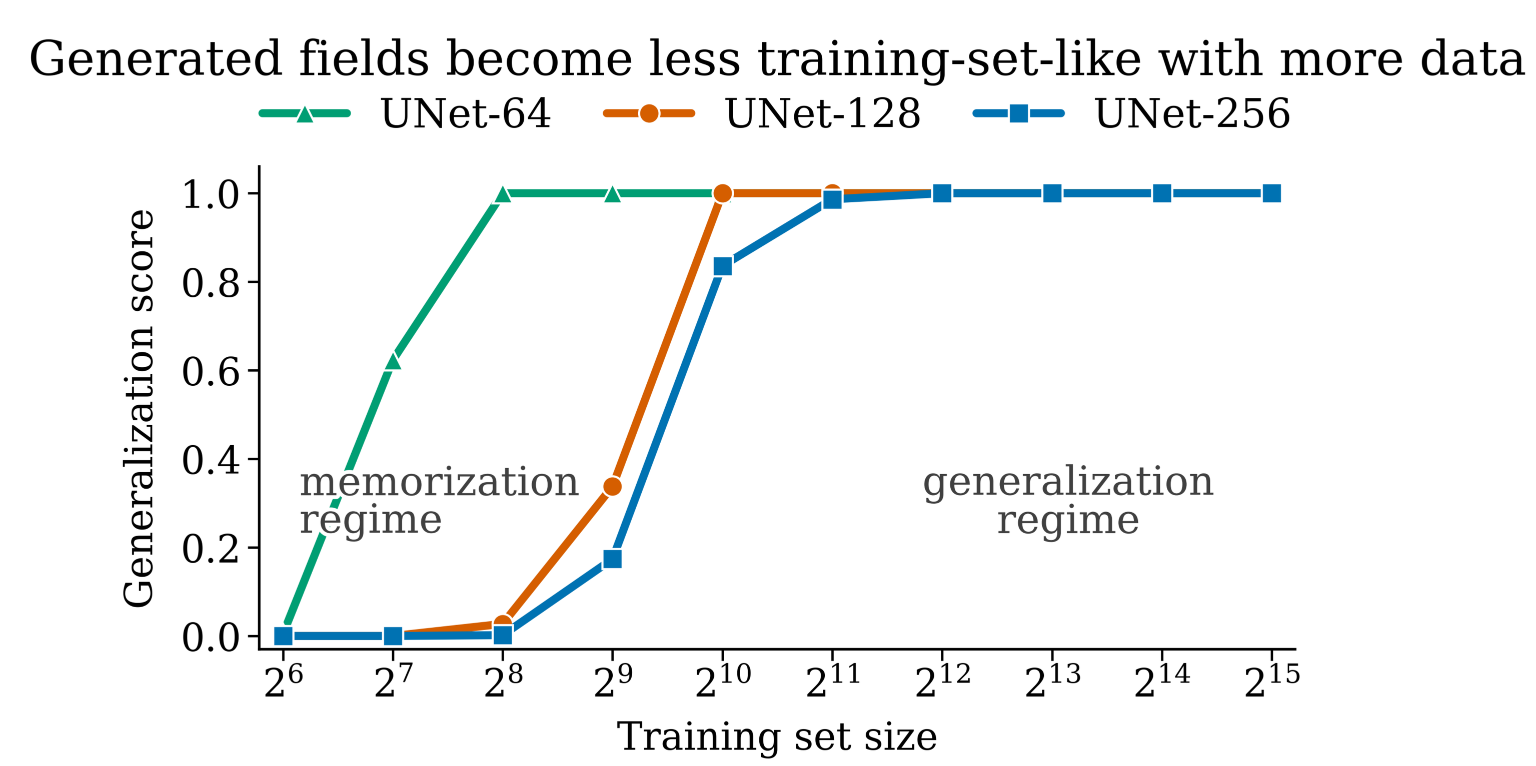
Calibration: does the generator obey θ?

Ask the conditional model for **held-out samples**, then read the cosmology back from its generated maps with a **frozen encoder**. If the conditioning is used, changing $\boldsymbol{\theta}_{\mathrm{ask}}$ should move the recovered $\boldsymbol{\theta}_{\mathrm{rec}}$ rather than leaving it near the training-set average.



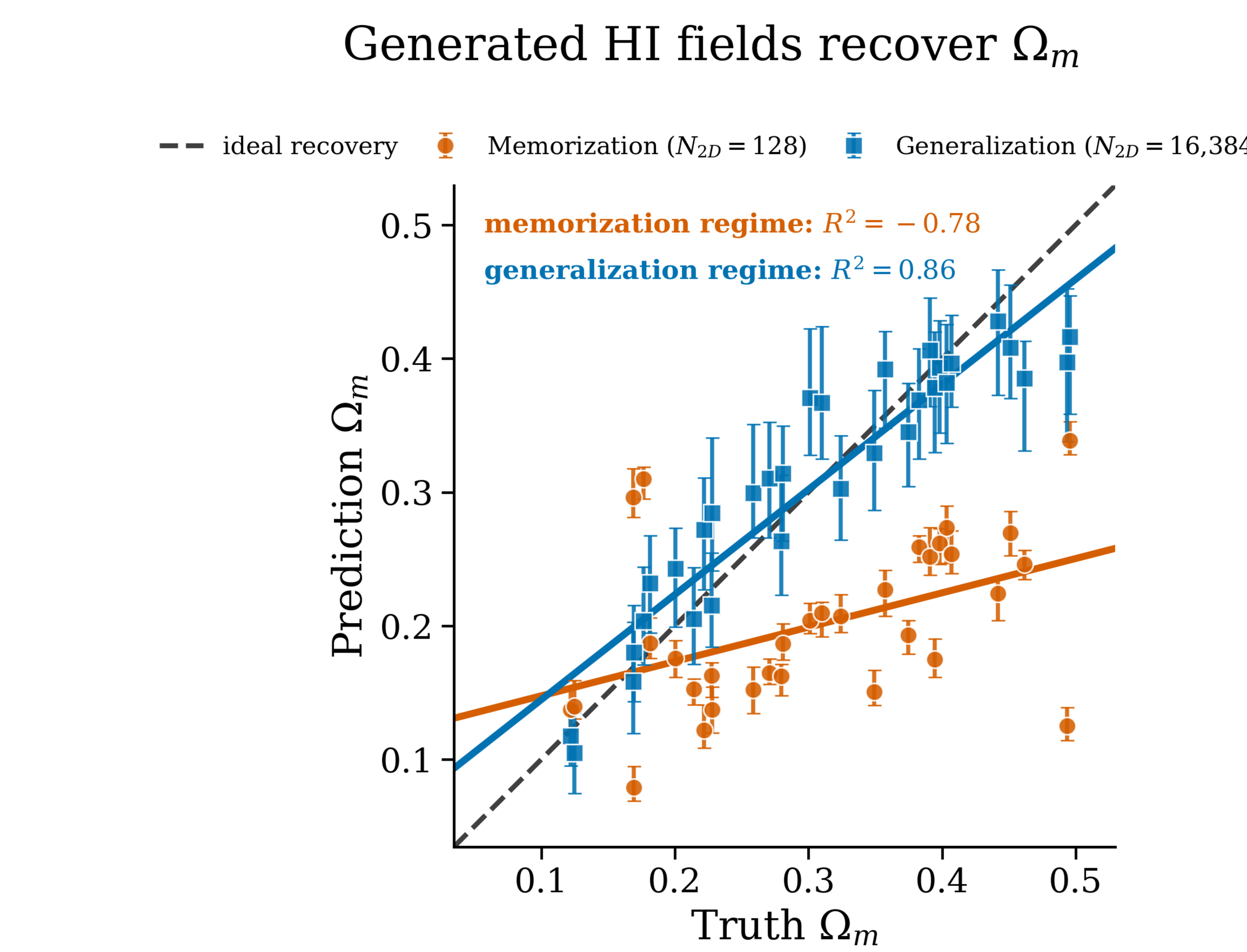
Encoder setup: VGG16 is frozen (ImageNet); only the regression head is trained on *real* slices from non-held-out simulations. Held-out test simulations are used only for encoder validation and generator calibration.

Memorization → generalization transition



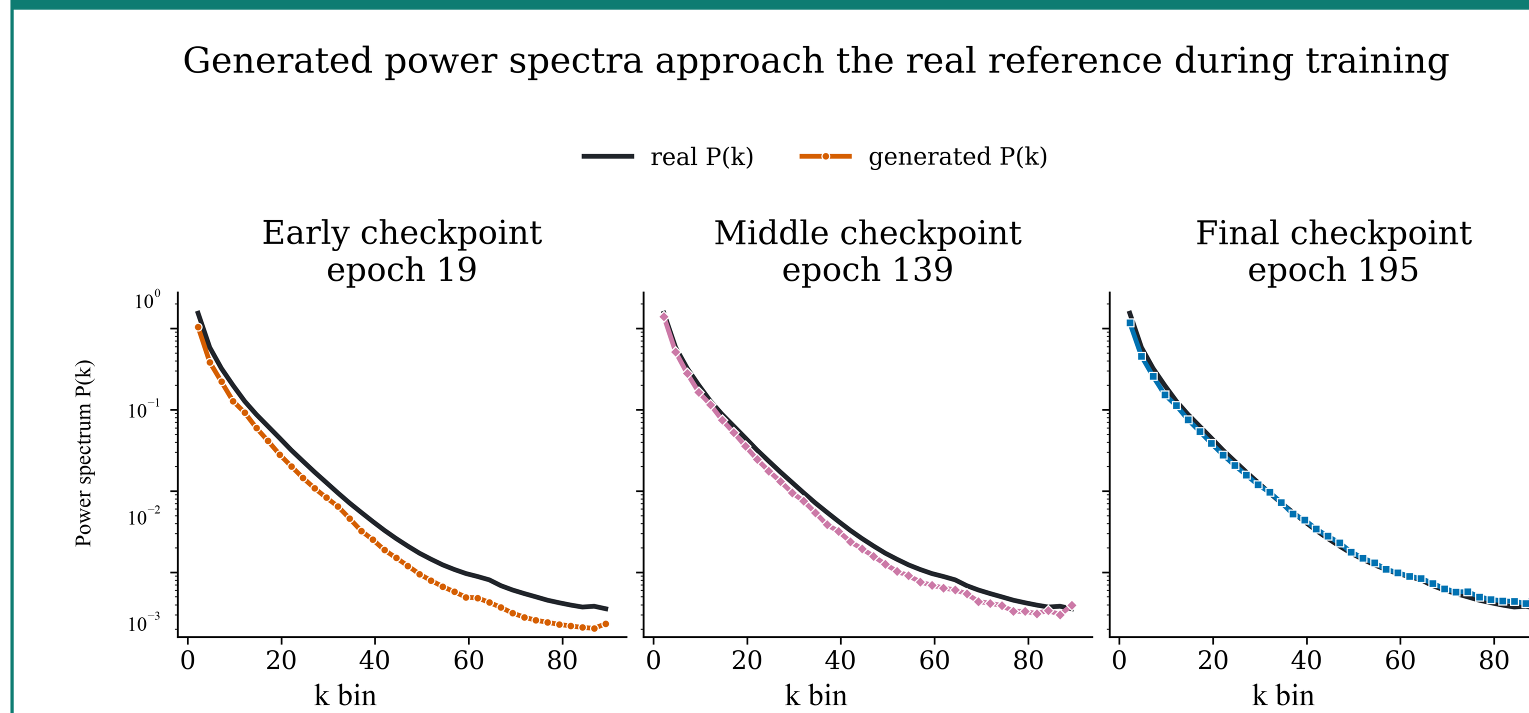
- Score uses a **q95-calibrated threshold** (τ = 95th pct train–train NN similarity): higher means generated fields are less training-set-like in PCA space.
- **Small** N_{2D}: generated maps remain close to training fields.
- **Larger** N_{2D}: UNet-64 / 128 / 256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Calibration results



Main result: Ω_m recovery is much flatter under memorization (**slope 0.26**, $N_{2D} = 2^7$) than in the high-data regime (**slope 0.79**, $N_{2D} = 2^{14}$). Error bars are the 16–84 percentile spread over generated seeds at fixed $\boldsymbol{\theta}_{\mathrm{ask}}$. We lead with Ω_m because the real-held-out VGG probe is strongest there ($R^2 = 0.91$; $\sigma_8 = 0.74$, feedback parameters weaker).

Physical fidelity



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power over many k bins; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

1. **Memorization is detectable.** Nearest-neighbor and reproducibility tests locate a data-size transition in CAMELS HI maps.
2. **Conditioning is recoverable.** High-data generated fields track $\boldsymbol{\theta}_{\mathrm{ask}}$ much better than memorization-regime fields.
3. **Fidelity requires generalization.** Useful emulators must be novel, statistically faithful, and calibrated before inference use.